Prediction.R

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```
#Prediction
# [a]
myData <- read.csv("marketing_sales_data.csv")
dim(myData)
```

```
## [1] 1000    7
```

```
str(myData)
```

```
## 'data.frame':    1000 obs. of  7 variables:
##  $ sales  : num  18100 19707 12816 13175 10668 ...
##  $ spend  : num  1935 4758 3687 3033 864 ...
##  $ price  : num  35.9 85.9 132.2 112.5 122.9 ...
##  $ clicks : int  9485 5048 212 2024 3341 8511 681 955 2064 9883 ...
##  $ promo  : num  33.12 5.93 14.44 19.89 45.98 ...
##  $ emails : int  1568 754 1945 1632 1918 368 1876 514 1368 1115 ...
##  $ reviews: num  3.2 3.1 5 1.8 2 1.5 4.7 4.1 4.3 4.6 ...
```

```
sum(is.na(myData))
```

```
## [1] 0
```

The dataset contains 1,000 observations and 7 variables, with absolutely zero missing valu es present.

```
# [b]
set.seed(003)
training_index <- sample(x = nrow(myData),
                                size = 0.70 * nrow(myData),
                                replace = FALSE)

training_data <- myData[training_index, ]
validation_data <- myData[-training_index, ]

dim(training_data)
```

```
## [1] 700   7
```

```
dim(validation_data)
```

```
## [1] 300   7
```

The data was partitioned into a 70% training set (700 observations) and a 30% validation se t (300 observations) using a random seed (003) to ensure reproducibility.

```r
# [c]
Model_1 <- lm(sales ~ spend + price + clicks + promo + emails + reviews,
                    data = training_data)
summary(Model_1)
```

```
##
## Call:
## lm(formula = sales ~ spend + price + clicks + promo + emails +
##     reviews, data = training_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1472.97  -342.16    12.78   328.11  1413.64
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)  4.899e+03  9.245e+01  52.990   <2e-16 ***
## spend        2.513e+00  1.319e-02 190.464   <2e-16 ***
## price       -1.500e+01  4.674e-01 -32.086   <2e-16 ***
## clicks       7.974e-01  6.698e-03 119.054   <2e-16 ***
## promo        4.709e+01  1.315e+00  35.802   <2e-16 ***
## emails      -8.311e-03  3.439e-02  -0.242    0.809
## reviews      9.387e+00  1.651e+01   0.569    0.570
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 501.3 on 693 degrees of freedom
## Multiple R-squared:  0.9864, Adjusted R-squared:  0.9862
## F-statistic:  8353 on 6 and 693 DF,  p-value: < 2.2e-16
```

```r
flextable::as_flextable(Model_1)
```

	Estimate	Standard Error	t value	Pr(>|t|)	
(Intercept)	4,899.113	92.454	52.990	0.0000	***
spend	2.513	0.013	190.464	0.0000	***
price	-14.998	0.467	-32.086	0.0000	***
clicks	0.797	0.007	119.054	0.0000	***
promo	47.095	1.315	35.802	0.0000	***
emails	-0.008	0.034	-0.242	0.8091	
reviews	9.387	16.506	0.569	0.5697	

*Signif. codes: 0 <= '***' < 0.001 < '**' < 0.01 < '*' < 0.05*

	Estimate	Standard Error	t value	Pr(>|t|)

Residual standard error: 501.3 on 693 degrees of freedom

Multiple R-squared: 0.9864, Adjusted R-squared: 0.9862

F-statistic: 8353 on 693 and 6 DF, p-value: 0.0000

The initial regression model (Model_1) using all predictors yielded an adjusted R^2 of 0.9862, but the emails and reviews variables were not statistically significant.

```
# [d]
Model_2 <- lm(sales ~ spend + price + clicks + promo,
                    data = training_data)
summary(Model_2)

##
## Call:
## lm(formula = sales ~ spend + price + clicks + promo, data = training_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1488.46  -338.44    12.09   332.86  1412.22
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)  4.917e+03  7.090e+01   69.36   <2e-16 ***
## spend        2.513e+00  1.317e-02  190.71   <2e-16 ***
## price       -1.499e+01  4.658e-01  -32.17   <2e-16 ***
## clicks       7.973e-01  6.689e-03  119.21   <2e-16 ***
## promo        4.710e+01  1.314e+00   35.85   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 500.7 on 695 degrees of freedom
## Multiple R-squared:  0.9864, Adjusted R-squared:  0.9863
## F-statistic: 1.256e+04 on 4 and 695 DF,  p-value: < 2.2e-16

flextable::as_flextable(Model_2)
```

	Estimate	Standard Error	t value	Pr(>|t|)	
(Intercept)	4,917.313	70.898	69.358	0.0000	***
spend	2.513	0.013	190.713	0.0000	***

	Estimate	Standard Error	t value	Pr(>|t|)	
price	-14.985	0.466	-32.173	0.0000	***
clicks	0.797	0.007	119.207	0.0000	***
promo	47.105	1.314	35.855	0.0000	***

*Signif. codes: 0 <= '***' < 0.001 < '**' < 0.01 < '*' < 0.05*

Residual standard error: 500.7 on 695 degrees of freedom

Multiple R-squared: 0.9864, Adjusted R-squared: 0.9863

F-statistic: 1.256e+04 on 695 and 4 DF, p-value: 0.0000

The refined regression model (Model_2) excluded the insignificant variables, yielding a slightly improved adjusted R^2 of 0.9863 with all remaining predictors being highly significant.

```r
# [e]
source("hamR.R")

predicted_Model_1 <- predict(Model_1, newdata = validation_data)
predicted_Model_2 <- predict(Model_2, newdata = validation_data)

performance_Model_1 <- calculate_metrics_prediction(actual = validation_data$sales,
                                                    predicted = predicted_Model_1)
performance_Model_2 <- calculate_metrics_prediction(actual = validation_data$sales,
                                                    predicted = predicted_Model_2)

data.frame(Model_1 = performance_Model_1,
           Model_2 = performance_Model_2) |> round(4)
```

```
##       Model_1  Model_2
## RMSE 480.7540 480.2726
## ME   -10.1136  -9.2802
## MAD  382.4330 382.1370
## MPE   -0.1635  -0.1591
```

I recommend deploying Model_2 because it achieved a lower RMSE (480.2726) and MAD (382.1370) on the validation data, offering better predictive accuracy with a simpler model.

Classification.R

Muhammad Shafiq

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```
#Classification Problem
#[a]
myData <- read.csv("marketing_conversion_data_v2.csv")
dim(myData)
```

```
## [1] 1000    7
```

```
str(myData)
```

```
## 'data.frame':    1000 obs. of  7 variables:
##  $ spend    : num  57.5 47.9 59.7 72.8 46.5 ...
##  $ clicks   : int  18 16 12 9 15 14 16 15 16 10 ...
##  $ discount : num  19.2 7.3 17.2 8.9 33.3 38.2 7.5 38.4 23.3 15.5 ...
##  $ age      : int  41 18 31 38 37 62 38 25 34 18 ...
##  $ email_len: int  15 50 24 57 31 37 21 58 21 58 ...
##  $ tenure   : int  29 33 27 20 31 32 28 14 27 38 ...
##  $ convert  : int  1 1 1 1 0 0 1 1 1 1 ...
```

```
sum(is.na(myData))
```

```
## [1] 0
```

The dataset contains 1,000 observations and 7 variables, with no missing values detected.

```
#[b]
set.seed(003)
training_index <- sample(x = nrow(myData),
                         size = 0.70 * nrow(myData),
                         replace = FALSE)

training_data <- myData[training_index, ]
validation_data <- myData[-training_index, ]

dim(training_data)
```

```
## [1] 700   7
```

```
dim(validation_data)
```

```
## [1] 300   7
```

The data was partitioned into a 70% training set (700 observations) and a 30% validation se t (300 observations) using a random seed (003) to ensure reproducibility.

```r
#[c]
Model_1<- glm(convert ~ spend + clicks + discount + age + email_len + tenure,
                    family = binomial,
                    data = training_data)

summary(Model_1)
```

```
##
## Call:
## glm(formula = convert ~ spend + clicks + discount + age + email_len +
##     tenure, family = binomial, data = training_data)
##
## Coefficients:
##               Estimate Std. Error z value Pr(>|z|)
## (Intercept) -4.466e+00  6.915e-01  -6.458 1.06e-10 ***
## spend        5.660e-02  7.217e-03   7.842 4.42e-15 ***
## clicks       2.774e-01  2.762e-02  10.042  < 2e-16 ***
## discount    -8.647e-02  1.010e-02  -8.564  < 2e-16 ***
## age          6.850e-03  9.172e-03   0.747    0.455
## email_len    8.148e-05  6.959e-03   0.012    0.991
## tenure       1.001e-02  8.030e-03   1.246    0.213
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 958.28  on 699  degrees of freedom
## Residual deviance: 720.73  on 693  degrees of freedom
## AIC: 734.73
##
## Number of Fisher Scoring iterations: 4
```

```r
flextable::as_flextable(Model_1)
```

	Estimate	Standard Error	z value	Pr(>|z|)	
(Intercept)	-4.466	0.691	-6.458	0.0000	***
spend	0.057	0.007	7.842	0.0000	***
clicks	0.277	0.028	10.042	0.0000	***
discount	-0.086	0.010	-8.564	0.0000	***
age	0.007	0.009	0.747	0.4551	
email_len	0.000	0.007	0.012	0.9907	
tenure	0.010	0.008	1.246	0.2127	

*Signif. codes: 0 <= '***' < 0.001 < '**' < 0.01 < '*' < 0.05*

	Estimate	Standard Error	z value	Pr(>|z|)

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 958.3 on 699 degrees of freedom

Residual deviance: 720.7 on 693 degrees of freedom

The initial logistic regression model (Model_1) yielded an AIC of 734.73. While spend, clicks, and discount were highly significant, the variables age, email_len, and tenure were not statistically significant.

```
# [d] Develop Model_2
Model_2 <- glm(convert ~ spend + clicks + discount,
                    family = binomial,
                    data = training_data)

summary(Model_2)

##
## Call:
## glm(formula = convert ~ spend + clicks + discount, family = binomial,
##     data = training_data)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept) -3.975251   0.528373  -7.524 5.33e-14 ***
## spend        0.056373   0.007183   7.848 4.24e-15 ***
## clicks       0.277358   0.027586  10.054  < 2e-16 ***
## discount    -0.086182   0.010071  -8.557  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 958.28  on 699  degrees of freedom
## Residual deviance: 722.78  on 696  degrees of freedom
## AIC: 730.78
##
## Number of Fisher Scoring iterations: 4
```

Yes, Model_2 was developed by dropping the insignificant variables to create a more parsimonious model. Model_2 yielded a lower (better) AIC of 730.78, with all remaining predictors (spend, clicks, discount) maintaining high statistical significance.

```r
# [e]
source("hamR.R")

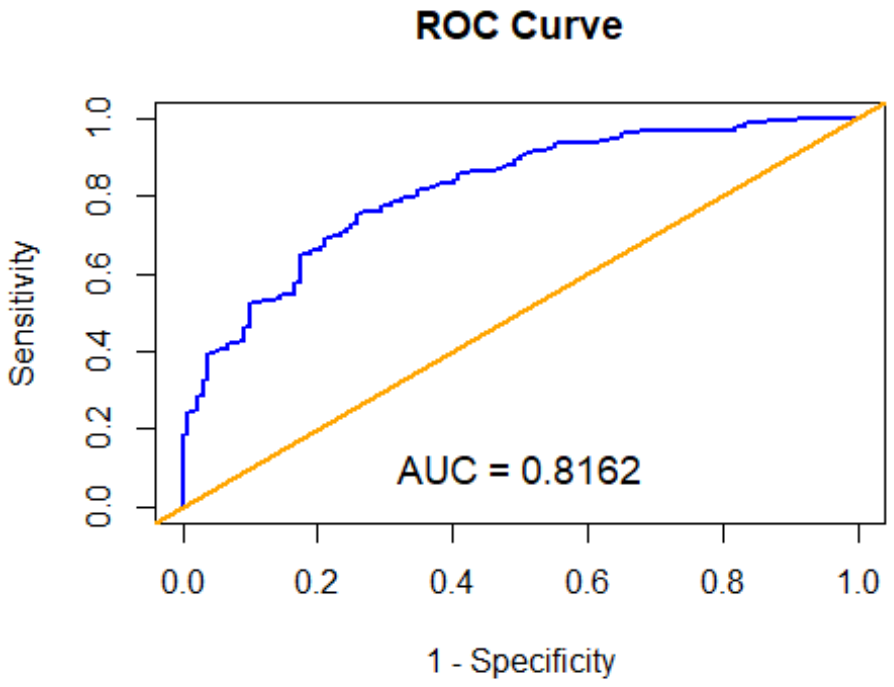
predicted_Model_1 <- predict(Model_1, newdata = validation_data, type = "resp
onse")

confusion_matrix(actual = validation_data$convert,
                 predicted = predicted_Model_1)
```

```
##       Predicted
## Actual  0   1
##      0 93  39
##      1 38 130
##                   Metric  Value
##  Misclassification Rate 0.2567
##                 Accuracy 0.7433
##              Sensitivity 0.7738
##                Precision 0.7692
##              Specificity 0.7045
```

```r
roc_curve(actual = validation_data$convert,
          predicted = predicted_Model_1)
```

```
## AUC: 0.8162
```



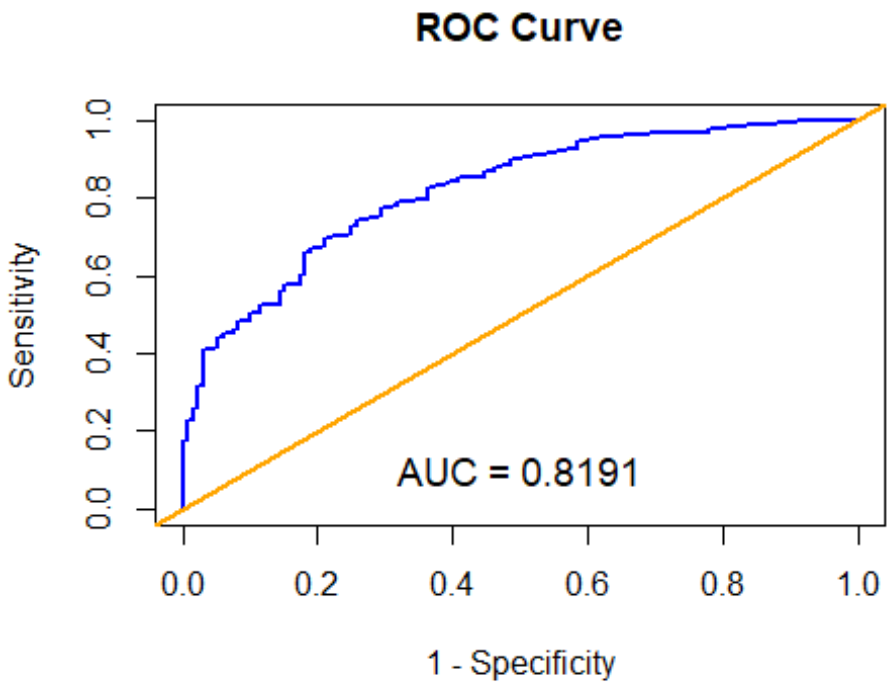

```r
predicted_Model_2 <- predict(Model_2, newdata = validation_data, type = "resp
onse")
```

```
confusion_matrix(actual = validation_data$convert,
                 predicted = predicted_Model_2)
```

```
##       Predicted
## Actual  0   1
##      0 93  39
##      1 38 130
##                  Metric  Value
##  Misclassification Rate 0.2567
##                Accuracy 0.7433
##             Sensitivity 0.7738
##               Precision 0.7692
##             Specificity 0.7045
```

```
roc_curve(actual = validation_data$convert,
          predicted = predicted_Model_2)
```

```
## AUC: 0.8191
```



Both models achieved the exact same accuracy (74.33%), sensitivity (77.38%), and specificity (70.45%) on the validation dataset. However, Model_2 achieved a slightly higher AUC (0.8191) compared to Model_1 (0.8162). Therefore, Model_2 is recommended because it is a simpler model that delivers slightly better overall discrimination capability.